

Sanjeev Krishnan Kamalamurugan

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OBJECTIVE: Seeking internship in Digital Design and Verification starting from January 2026.

EDUCATION

M.S., Electrical and Computer Engineering **GPA: 3.7 / 4.0**

Portland State University, Portland, Oregon (Expected Graduation: December 2026)

Coursework: Pre-Silicon Validation, Assertion Based Verification, Microprocessor System Design, Advanced Computer Architecture. **Courses by June 2026:** Post Silicon Validation, Formal Verification

B.E., Electronics and Communication Engineering **GPA: 8.3 / 10**

Anna University, Chennai, India (Graduated: April 2023)

TECHNICAL SKILLS

Programming & Scripting Languages	Verilog, SystemVerilog, C/C++, CUDA C++, TCL, Python
Hardware Concepts & Architectures	x86, RISC-V, MIPS, NVIDIA GPU Architecture, DDR DRAM, Cache, Virtual Memory, Branch Prediction, Pipelining, Out-of-order execution, MLP, SMT, Cache Prefetch
Verification Methods & Concepts	UVM, Class based Verification, SVA, Functional Coverage, Constraints and Randomization, Object Oriented Programming (OOP)
Protocols & Interfaces	AMBA, MESIF, UART, I2C, SPI
Tools & Operating System	Mentor QuestaSim, Synopsis VC Formal, Cadence JasperGold, Vivado, Vitis, Git, Visual Studio Code, Linux

PROJECTS

UVM-Based Arithmetic Floating Point Unit Verification[System Verilog]

Designed **IEEE 754-compliant Arithmetic FPU** in achieving **95% functional and 80% code coverage**. Generated **300+ stimulus** through constraints and randomization verifying corner cases like **Infinity, NaN, and subnormals** to ensure full compliance of the unit.

CNN based Hand Drawn Digit Recognizer [Verilog, C]

Designed and deployed a **hardware-accelerated CNN inference** system on **FPGA** with a custom **AXI Matrix Multiplication IP** for high-speed floating-point computation. Integrated **MicroBlaze firmware, UART protocol, and React UI** for real-time digit recognition with recognition **accuracy of over 95%**.

Assertion Based Verification of Asynchronous FIFO[System Verilog]

Designed and verified an **45-deep Asynchronous FIFO** with gray code pointers and synchronizers using **SystemVerilog Assertions**, achieving **100% coverage** through **system verilog cover properties**. Validated data **integrity, metastability, and boundary conditions** like data loss, timing hazards.

RISC-V RV32IM Simulator in C

Implemented **5-stage pipelined RISC-V ISA** functionality in single cycle model, addressing the **64KB memory constraint** managing the initialisation of **program counter and stack pointer**. Developed robust testcases in assembly language and generated **memory image files** using **RISC-V cross compiler**.

System Verilog Design and Verification of Cache Controller

Designed and verified **Last Level cache** featuring **16-way associativity** and **64-byte line size** using **write-back and write-allocate** policies. Implemented **P-LRU policy** and **MESI protocol** for eviction and coherence. Verified with comprehensive test cases achieving performance of **99.6% hit ratio**.

CERTIFICATION

Accelerated Computing in Modern CUDA C++ Issued by Nvidia · (July 2025)

Completed hands-on training in CUDA C++ for GPU-accelerated computing, covering parallel programming, memory optimization, kernel execution, thread hierarchy, and performance profiling.

Basic Static Timing Analysis Issued by Cadence Design Systems · (April 2025)

Covered concepts like **cell delay, net delay, timing paths, and timing checks**. Gained understanding of timing constraints and analysis techniques essential for **digital circuit timing verification**.